

Numerical Stiffness

- ❑ **Definition:** Consider a system which can be represented by a system of ODEs (generally as a nonlinear function).

The system of ODEs is called **numerically stiff** (or simply **stiff**) if it contains rapidly decaying (**fast**) dynamics alongside slowly varying (**slow**) dynamics.

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t) \\ \mathbf{x}(t_0) = \mathbf{x}_0 \in \mathcal{R}^m \end{cases}$$

- ❖ Even though the fast dynamics die out quickly, they constrain the numerical solver to take **very small step sizes** for stability, much smaller than what would be needed to solve for the slow dynamics.



- Simultaneous presence of widely separated time scales in the system

- Explicit solvers become **unstable** unless the step size is very small (so it's about numerical stability)

- ❖ **Example:**

$$\begin{cases} \frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1000 & 0 \\ 0 & -1000 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ \begin{bmatrix} x_1(0) & x_2(0) \end{bmatrix}^T = \begin{bmatrix} 10 & 10 \end{bmatrix}^T \end{cases}$$

Handwritten notes for the example:

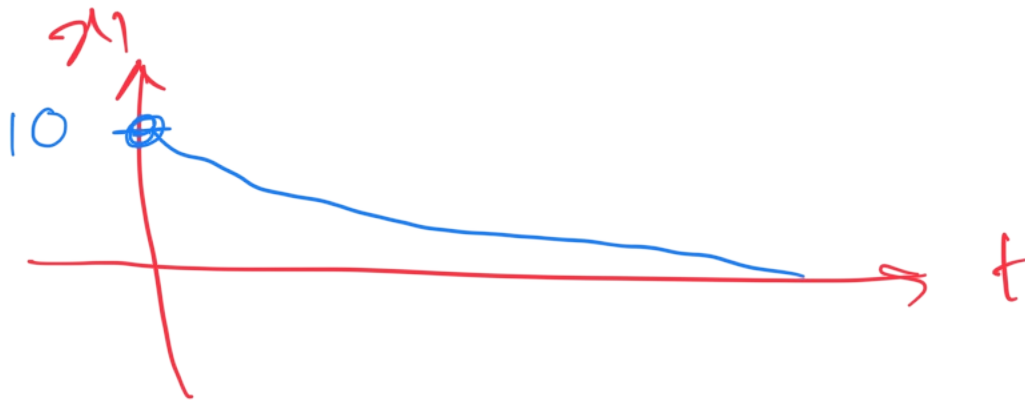
- $\begin{cases} dx_1/dt = -50x_1 \\ dx_2/dt = -1000x_2 \end{cases}$
- $\begin{cases} x_1 = 10e^{-50t} \\ x_2 = 10e^{-1000t} \end{cases}$

Recall:

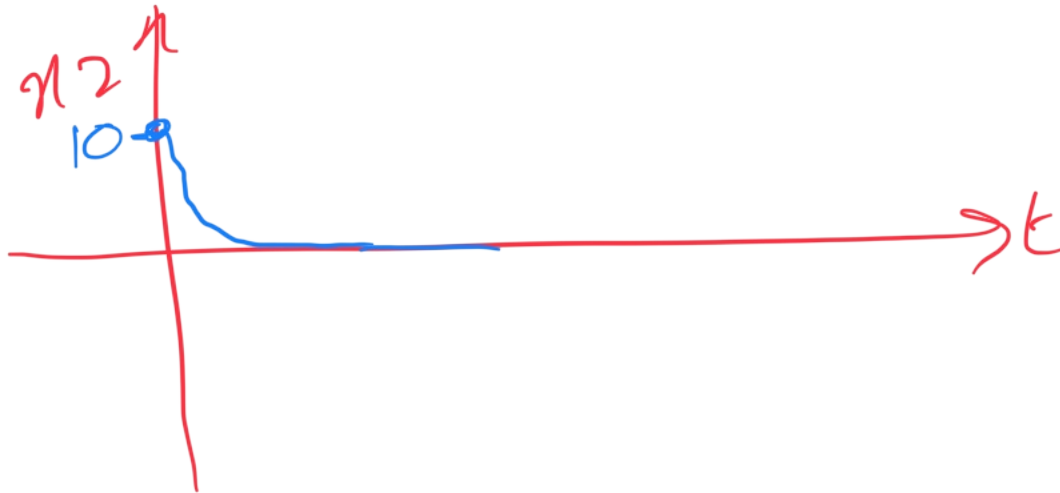
$$\frac{dx}{dt} = -ax \Rightarrow x = Ae^{-at}$$

Handwritten notes for the recall:

- $a > 0$
- $x(0) = Ae^0 = A$



$$x_1 = 10e^{-50t}$$



$$x_2 = 10e^{-1000t}$$

$$t \rightarrow \infty$$

$$\swarrow \searrow$$

$$x_1 \rightarrow 0 \quad x_2 \rightarrow 0$$

Numerical Stiffness

❖ Consider an LTI system:

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} \\ \mathbf{x}(t_0) = \mathbf{x}_0 \end{cases}$$

Solution:

$$\mathbf{x} = e^{\mathbf{A}t} \mathbf{x}_0$$

Taylor's Series:

$$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{(\mathbf{A}t)^2}{2} + \dots$$

$$e^{\mathbf{A}t} = \sum_{n=0}^{\infty} \frac{(\mathbf{A}t)^n}{n!}$$

➤ Suppose the matrix $\mathbf{A}$ can be expressed as $\mathbf{A} = \mathbf{S}\mathbf{\Lambda}\mathbf{S}^{-1}$ such that:

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_m \end{bmatrix}$$

$$\lambda_i \in \mathbb{C}$$

$\lambda_1, \lambda_2, \dots, \lambda_m$ are called the eigenvalues of the matrix $\mathbf{A}$

$$\det(\lambda_i \mathbf{I} - \mathbf{A}) = 0$$

❖ If a square matrix $\mathbf{A}$ acts as a linear transformation on a vector $\mathbf{x}$, in general, multiplying the matrix changes both the length and the direction of the vector. However, some special vectors keep their direction the same (they only get stretched, compressed, or flipped). These special vectors are eigenvectors, and the factors by which they are stretched are eigenvalues.

Numerical Stiffness

❖ Consider an LTI system:

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} \\ \mathbf{x}(t_0) \neq \mathbf{x}_0 \end{cases}$$



Solution: $\mathbf{x} = e^{\mathbf{A}t} \mathbf{x}_0$

Taylor's Series:

$$e^{\mathbf{A}t} = \sum_{n=0}^{\infty} \frac{(\mathbf{A}t)^n}{n!}$$

✓ assuming in the range of small time-steps, states do not change much so we can use linearized system

➤ Suppose the matrix $\mathbf{A}$ can be expressed as $\mathbf{A} = \mathbf{S}\mathbf{\Lambda}\mathbf{S}^{-1}$ such that:

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_m \end{bmatrix}$$

$$\lambda_i \in \mathbb{C}$$



$\lambda_1, \lambda_2, \dots, \lambda_m$ are called the eigenvalues of the matrix $\mathbf{A}$

$$\det(\lambda_i \mathbf{I} - \mathbf{A}) = 0$$



❖ $\mathbf{v}_i \in \mathbb{C}^m$ is an eigenvector of $\mathbf{A}$ if $(\lambda_i \mathbf{I} - \mathbf{A}) \mathbf{v}_i = 0$ or $\mathbf{A} \mathbf{v}_i = \lambda_i \mathbf{v}_i$

✓ It can also be proven that $\mathbf{S} = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_m]$

Numerical Stiffness

❖ Consider an LTI system:

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} \\ \mathbf{x}(t_0) = \mathbf{x}_0 \end{cases}$$

Solution:

$$\begin{aligned} \mathbf{x} &= e^{\mathbf{A}t} \mathbf{x}_0 \\ \mathbf{A} &= \mathbf{S}\mathbf{\Lambda}\mathbf{S}^{-1} \end{aligned} \quad \mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_m \end{bmatrix}$$

$$\mathbf{x}(t) = e^{\mathbf{A}t} \mathbf{x}_0 = e^{\mathbf{S}\mathbf{\Lambda}\mathbf{S}^{-1}t} \mathbf{x}_0 = \mathbf{S} e^{\mathbf{\Lambda}t} \mathbf{S}^{-1} \mathbf{x}_0$$

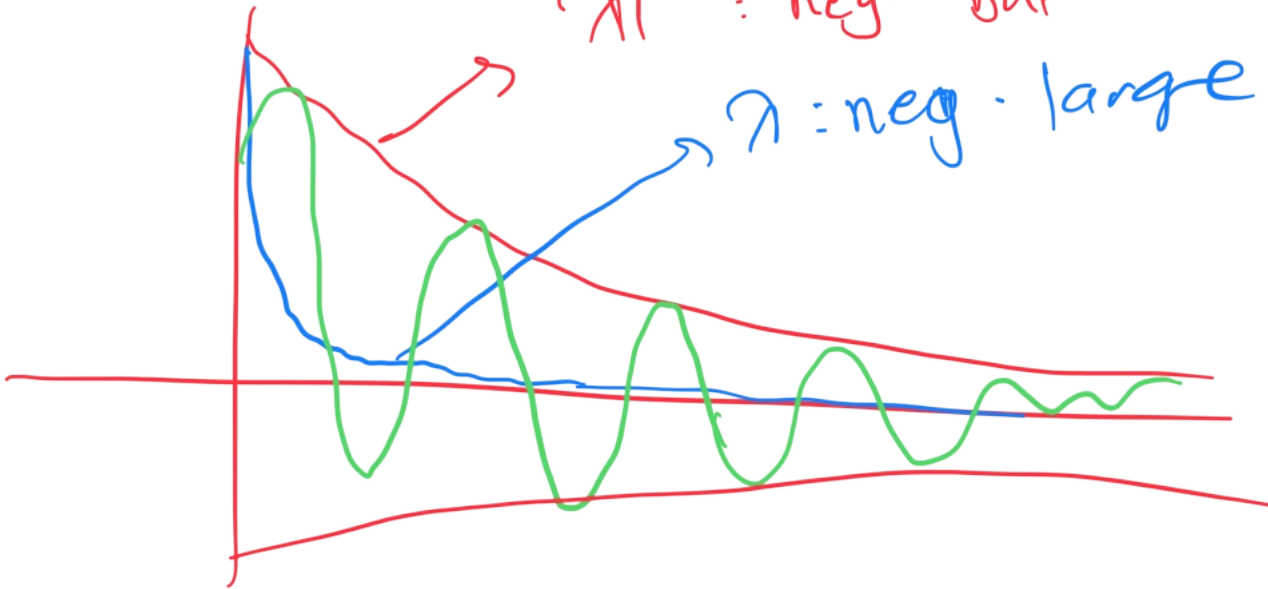
$$e^{\mathbf{\Lambda}t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & e^{\lambda_2 t} & \\ & & \ddots \\ & & & e^{\lambda_m t} \end{bmatrix}$$

$\lambda = \sigma + i\gamma$
 $\text{Re}\{\lambda_i\} < 0$

- ✓ $e^{\lambda_i t}$ defines the internal modes (increasing, decreasing)
- ✓ λ_i determines how the modes in $\mathbf{x}(t)$ change (fast, slow)

λ_1 : neg. but not large

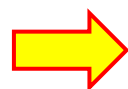
λ : neg. - large



Numerical Stiffness

❖ Consider an LTI system:

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} \\ \mathbf{x}(t_0) \neq \mathbf{x}_0 \end{cases}$$



Solution:

$$\mathbf{x} = \mathbf{S}e^{\mathbf{A}t}\mathbf{S}^{-1}\mathbf{x}_0$$

$$e^{\mathbf{A}t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & e^{\lambda_2 t} & \\ & & e^{\lambda_m t} \end{bmatrix}$$

- $$\begin{cases} \checkmark e^{\lambda_i t} & \text{defines the internal modes (increasing, decreasing)} \\ \checkmark \lambda_i & \text{determines how the modes in } \mathbf{x}(t) \text{ change (fast, slow)} \end{cases}$$

❖ **Stability of the LTI System:**

- If for any eigenvalue $\text{Re}(\lambda_i) > 0$, the term $e^{\lambda_i t}$ grows exponentially in time and the (actual) system is unstable
- ✓ Let's assume that we are simulating a stable system, i.e., all eigenvalues have $\text{Re}(\lambda_i) < 0$

Numerical Stiffness

❖ Consider an LTI system:

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} \\ \mathbf{x}(t_0) \neq \mathbf{x}_0 \end{cases}$$

➡ Solution:

$$\lambda_2 = -1000$$

$$\lambda_1 = -1 \pm 2i$$

$$\mathbf{x} = \mathbf{S}e^{\mathbf{A}t}\mathbf{S}^{-1}\mathbf{x}_0$$

$$e^{\mathbf{A}t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & e^{\lambda_2 t} & \\ & & e^{\lambda_m t} \end{bmatrix}$$

- ✓ $e^{\lambda_i t}$ defines the internal modes (increasing, decreasing)
- ✓ λ_i determines how the modes in $\mathbf{x}(t)$ change (fast, slow)

❖ Stiffness Ratio: indicates how much different fast and slow dynamics are!

$$\underbrace{|Re(\lambda_{\min})|}_{\text{very slow}} \ll \underbrace{|Re(\lambda_{\max})|}_{\text{very fast}} \quad \text{or} \quad \underbrace{S = \frac{|Re(\lambda_{\max})|}{|Re(\lambda_{\min})|}}_{\text{stiff}} = 10^3 \dots 10^{10}$$

➤ Stiff systems are harder to solve due to numerical instability!

Numerical Stability

- ❖ A numerically stable method ensures that the small errors (round-off and truncation) do not grow uncontrollably as the algorithm progresses.

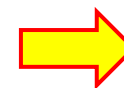
- ❖ Consider an LTI system:

Example of scalar state:

$$\begin{cases} \frac{dx}{dt} = ax \\ x_0 = 1 \end{cases}$$

$$a < 0$$

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} \\ \mathbf{x}(t_0) = \mathbf{x}_0 \end{cases}$$



Solution:

$$\mathbf{x} = e^{\mathbf{A}t} \mathbf{x}_0$$

Solution is known analytically:

$$x(t) = e^{at}$$

$$\begin{cases} t \rightarrow \infty \\ x \rightarrow \end{cases}$$

Let's use FE numerical method:

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \times \mathbf{f}(\mathbf{x}_n, t_n)$$

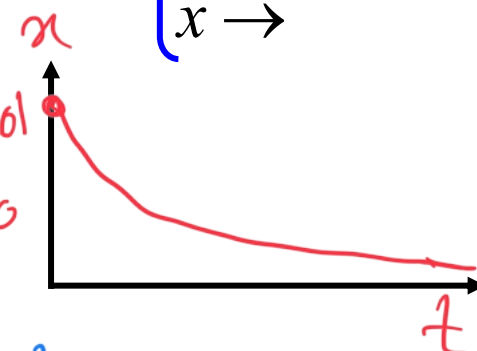
$$x_{n+1} = x_n + h a x_n = (1 + h a) x_n$$

$$x_{n+1} = (1 + h a)^n x_0$$

$$|1 - h \times 1000| < 1$$

$$\begin{cases} x_1 = (1 + h a) x_0 \\ x_2 = (1 + h a)^2 x_0 \\ \vdots \end{cases}$$

$$|1 + h a| < 1$$



- Numerical Stability requirement?

Numerical Stability

- ❖ Consider an LTI system solved with FE method:

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \times \mathbf{f}(\mathbf{x}_n, t_n)$$

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} \\ \mathbf{x}(t_0) = \mathbf{x}_0 \end{cases}$$



Solution:

$$\mathbf{x} = e^{\mathbf{A}t} \mathbf{x}_0$$

$$\mathbf{A} \in \mathbb{R}^{m \times m}, \text{Re}(\lambda_i) < 0$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h\mathbf{A}\mathbf{x}_n = (\mathbf{I} + h\mathbf{A})\mathbf{x}_n$$

$$\mathbf{x}_1 = (\mathbf{I} + h\mathbf{A})\mathbf{x}_0$$

$$\mathbf{x}_2 = (\mathbf{I} + h\mathbf{A})^2 \mathbf{x}_0$$

$$\vdots$$

$$\mathbf{x}_n = (\mathbf{I} + h\mathbf{A})^n \mathbf{x}_0$$

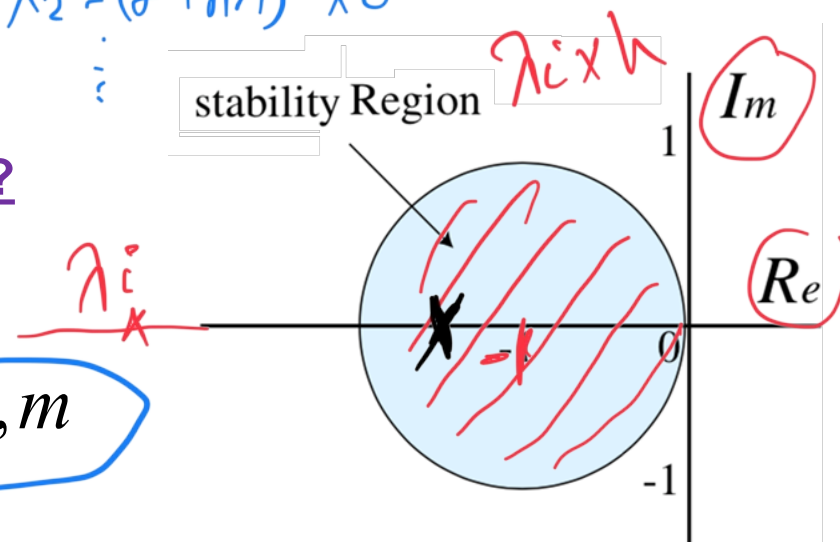
- Numerical Stability requirement?

$$\|\mathbf{I} + h\mathbf{A}\| < 1$$

$$\star \quad \|1 + h\lambda_i\| < 1, \text{ for } i = 1, 2, \dots, m$$

- What if one λ_i is very large negative?

very small h !



Numerical Stability

- ❖ Consider an LTI system solved with **BE** method:

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \times \mathbf{f}(\mathbf{x}_{n+1}, t_{n+1})$$

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} \\ \mathbf{x}(t_0) \neq \mathbf{x}_0 \end{cases}$$



Solution:

$$\mathbf{x} = e^{\mathbf{A}t} \mathbf{x}_0$$

$$\mathbf{A} \in \mathbb{R}^{m \times m}, \text{Re}(\lambda_i) < 0$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h \mathbf{A} \mathbf{x}_{n+1} \rightarrow (\mathbf{I} - h\mathbf{A}) \mathbf{x}_{n+1} = \mathbf{x}_n$$

$$\mathbf{x}_{n+1} = (\mathbf{I} - h\mathbf{A})^{-1} \mathbf{x}_n$$

$$\mathbf{x}_n = \left([\mathbf{I} - h\mathbf{A}]^{-1} \right)^n \mathbf{x}_0$$

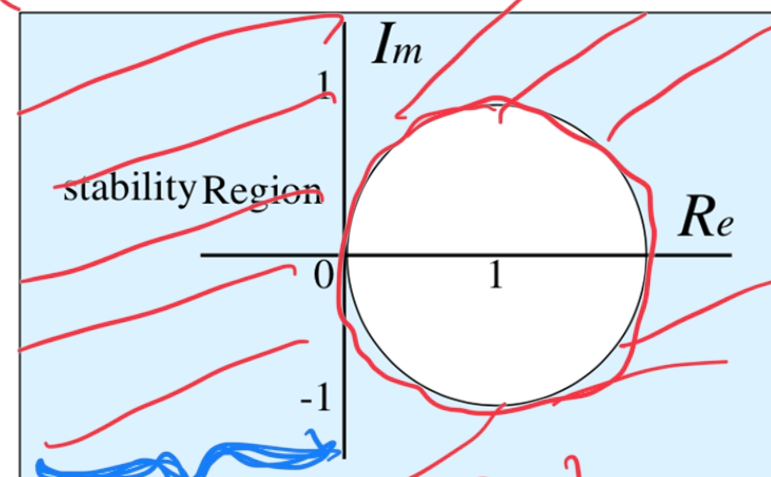
- Numerical Stability requirement?

$$\| [\mathbf{I} - h\mathbf{A}]^{-1} \| < 1 \quad \Rightarrow$$

$$\| 1 - h\lambda_i \| > 1, \text{ for } i = 1, 2, \dots, m$$

- What if one λ_i is very large negative?

no problem!



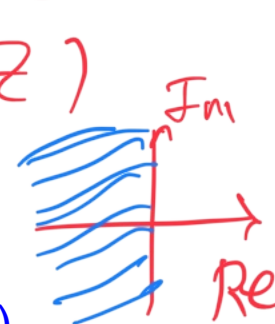
Numerical Stability

□ Important Definitions of Numerical Stability:

- Numerical stability is defined with respect to the LTI test system $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$
- Consider the difference equation $\mathbf{x}_{n+1} = \mathbf{x}_n + h \times \Phi(\mathbf{x}, t, h)$
- Consider the solution of the difference equation as $\mathbf{x}_{n+1} = \varphi(h\mathbf{A})\mathbf{x}_n$
- Define stability function $\varphi(z)$ where $z = h\mathbf{A}$

❖ A-Stability:

Method Φ is A-stable if $\|\varphi(z)\| < 1$ for all z where $\text{Re}(z) < 0$
(i.e., if its stability region contains the entire left-half of complex plane)



➡ ✓ the numerical method is stable for *any* step size

❖ L-Stability:

Method Φ is L-stable if it is A-stable and $\|\varphi(z)\| \rightarrow 0$ if $\text{Re}(z) \rightarrow -\infty$
(for very stiff problems (z large negative), the numerical solution damps to zero)

➡ ✓ effectively kills off very stiff modes (avoids numerical oscillations)

Numerical Stability

❖ Consider an LTI system solved with **TP** method:

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \frac{h}{2} \times (\mathbf{f}(\mathbf{x}_n, t_n) + \mathbf{f}(\mathbf{x}_{n+1}, t_{n+1}))$$

$$\mathbf{x}_{n+1} =$$

$$\mathbf{x}_{n+1} = \left(\mathbf{I} - \frac{1}{2} h \mathbf{A} \right)^{-1} \left(\mathbf{I} + \frac{1}{2} h \mathbf{A} \right) \mathbf{x}_n$$

➤ Stability function:

$$\varphi(z) =$$

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} \\ \mathbf{x}(t_0) \neq \mathbf{x}_0 \end{cases}$$



Solution:

$$\mathbf{x} = e^{\mathbf{A}t} \mathbf{x}_0$$

$$\mathbf{A} \in \Re^{m \times m}, \text{Re}(\lambda_i) < 0$$

Stability function:

$$\mathbf{x}_{n+1} = \varphi(h\mathbf{A}) \mathbf{x}_n$$

$$z = h\mathbf{A}$$

Numerical Stability

❖ Consider an LTI system solved with **TP** method:

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \frac{h}{2} \times (\mathbf{f}(\mathbf{x}_n, t_n) + \mathbf{f}(\mathbf{x}_{n+1}, t_{n+1}))$$

➤ Stability function: $\varphi(z) = \frac{1 + \frac{1}{2}z}{1 - \frac{1}{2}z}$

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x} \quad \mathbf{x}(t_0) \neq \mathbf{x}_0$$

❖ **A-Stability:**

$$\|\varphi(z)\| < 1 \quad \text{if} \quad \text{Re}(z) < 0$$

❖ **L-Stability:** A-stable and

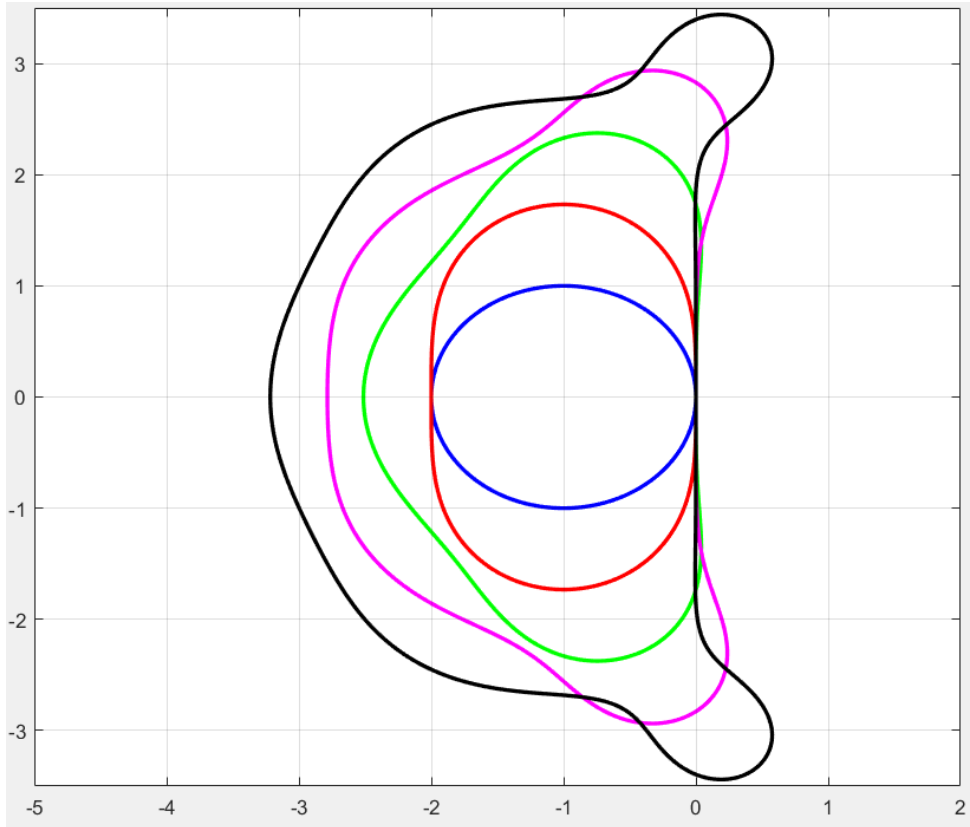
$$\|\varphi(z)\| \rightarrow 0 \quad \text{if} \quad \text{Re}(z) \rightarrow -\infty$$

❖ What about **BE** method?

$$\mathbf{x}_n = \left([\mathbf{I} - h\mathbf{A}]^{-1} \right) \mathbf{x}_n$$

Numerical Stability

❖ Stability Regions of Explicit RK Methods:



✓ Derive the stability function using the difference equation and the update formula and by replacing

$$z = h\mathbf{A}$$

$$\varphi(z) = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + \frac{z^5}{5!}$$

$$\Rightarrow |\varphi(z)| < 1$$

[Mathworks Link](#)

$$z = x + iy$$

$$\left| 1 + z + \frac{(z)^2}{2} + \frac{(z)^3}{6} + \frac{(z)^4}{24} + \frac{(z)^5}{120} \right| < 1$$

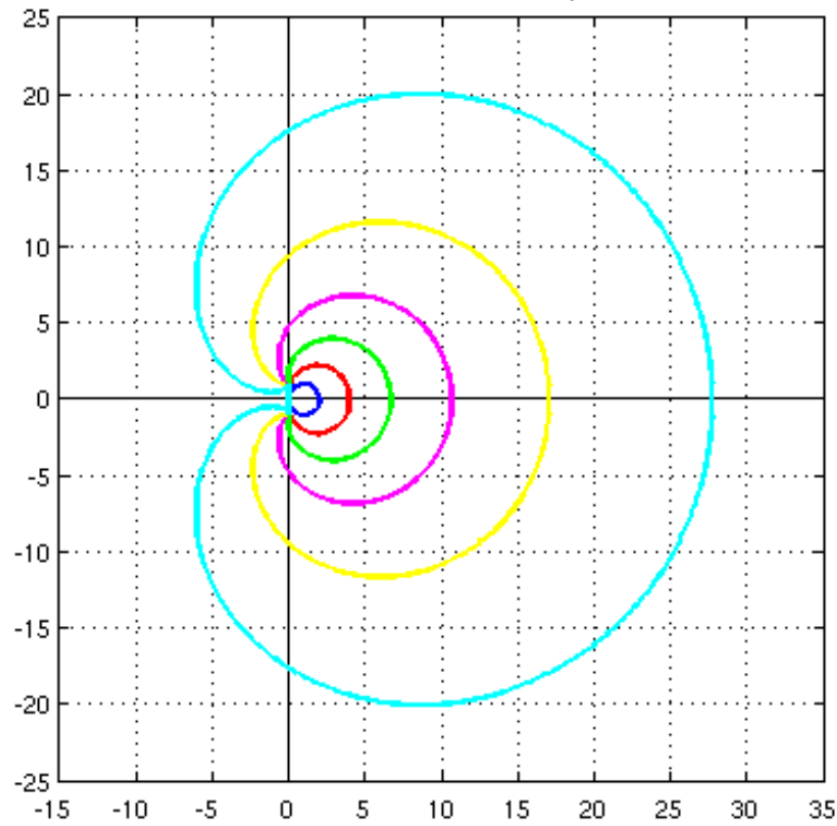
Numerical Stability

❖ Stability Regions of Backward Differentiation Formula (BDF):

✓ exteriors of the curves!

✓ Stability region decreases with order!

Backward differentiation orders 1-6 (exteriors of curves)



✓ Is it suitable for stiff systems?

[Mathworks Link](#)

Step-Size Control

To be continued...